

Subhadeep Ghorai

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PROFESSIONAL SUMMARY

Software Engineer and **Instructor** at Newton School of Technology building **AI-agent systems on the MERN stack**, with a parallel track as a **System Design educator to 300+ engineers (4.44/5 rating, 91.55% CSAT)**. Architected and shipped multi-pipeline AI-agent grading platforms (SESD-Agent) that replaced dozens of manual grading hours a week, and delivered 100+ System Design, OOP, and Software Engineering lectures across 3 cohorts. **2,500+ live coding sessions** mentoring 100+ students across 10+ countries. Backend-leaning full-stack engineer with working exposure to **Cloud (AWS, Firebase), Docker, Kubernetes, and CI/CD**, seeking SDE-1 / Backend / Full Stack roles where scalable system design and the ability to teach it are both valued.

TECHNICAL SKILLS

Languages	Java, Python, JavaScript (ES6+), TypeScript
Frontend	React.js, Redux, Tailwind CSS, Framer Motion, HTML5, CSS3
Backend	Node.js, Express.js, RESTful API Design, Socket.IO, Claude API / LLM Integration
Databases	MongoDB, MySQL, Firebase Firestore
Cloud	AWS, Firebase, Vercel
DevOps & CI/CD	Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Nginx, Linux, Git & GitHub
Testing & Tools	Postman, Playwright, axe-core (accessibility audits)
Architecture	System Design, API Design, Authentication & RBAC, OOP, Design Patterns, Microservices Basics
Core CS	Data Structures & Algorithms, Operating Systems, DBMS, Computer Organization & Architecture, Software Engineering

WORK EXPERIENCE

Instructor & Software Engineer — Newton School of Technology Dec 2025 – Present

- **Architected and deployed** AI-agent evaluation pipelines (RepoScore-V2, WAP-Agent) on the **MERN stack**, automating assessment workflows that previously cost dozens of manual hours per week.
- **Built** internal full-stack tools spanning project-evaluation pipelines, learning platforms, and operational dashboards, improving evaluation throughput for engineering cohorts.
- **Delivered** 100+ lectures on System Design, OOP, Software Engineering, and Design Patterns across 3 engineering cohorts, mentoring **300+ aspiring engineers** to a **91.55% CSAT** and **4.44/5 rating**.
- **Reviewed** 150+ GitHub pull requests and **resolved** 99+ technical grievances at scale without slowing cohort delivery.
- **Conducted** 30+ mock interviews, translating engineering hiring bar into feedback students could act on.

Teacher Mentor & Senior Coding Instructor — Codingal Inc. Dec 2023 – Jan 2026

- **Mentored** 100+ students across 10+ countries through **2,500+ live coding sessions** in Java, Python, AI, and Web Development, sustaining a **4.44/5** average rating.
- **Achieved** an **80% student renewal rate** by designing personalized, outcome-driven curricula and maintaining consistent session quality.
- **Trained** newly onboarded instructors and **evaluated** teaching and communication skills as Teacher Mentor, raising onboarding quality across the instructor pool.

TECHNICAL PROJECTS

SESD-Agent – AI-Agent Grading Platform (Flagship) [GitHub](#)

- *Problem:* Manually grading GitHub repos, case studies, and coding-activity streaks for 300+ engineers across cohorts would consume dozens of hours weekly.
- **Architected** three independent AI-agent pipelines on **Node.js**, integrating the **Claude API**, GitHub API, Google Sheets API, LeetCode API, and Codeforces API into a single grading system.
- **Automated** project evaluation by having Claude parse required UML docs and score backend/frontend quality, **grading 441 students in ~110 minutes** versus dozens of manual hours.
- **Built** a case-study evaluator that scores research depth, clarity, and real-world impact via Claude, **processing 154 submissions in ~40 minutes**.
- **Engineered** a zero-cost streak evaluator pulling live GitHub/LeetCode/Codeforces contribution data, **scoring each student in ~500ms with no LLM or API cost**.

Jana Academy – Full-Stack Learning Platform

GitHub | Live

- **Built** a live academy platform with **Google OAuth** authentication, a real-time Quiz Arena, and a role-based admin dashboard on **React** and Firebase Auth.
- **Instrumented** the platform with Vercel Web Analytics and Core Web Vitals monitoring, sustaining a **1.2s LCP** and **18ms FID** in production.
- **Scaled** the platform to **1,143 page views** and **117 unique visitors** across **45 registered users** and 5 active admins.

SmartReviewX – Project Review Automation

GitHub

- **Automated** the project-review workflow with a modular **OOP** architecture and real-time **WebSocket** updates, **cutting evaluation time by 95%** (3 minutes to 6–7 seconds per project).
- **Reduced recurring bugs by 25–30%** through structured modularization, simplifying long-term maintenance.

LifeDrop – Full-Stack Blood Donation Platform

GitHub | Live

- **Developed** a real-time platform connecting donors, recipients, and hospitals with **RBAC dashboards** and instant WhatsApp alerts via Twilio.
- **Implemented** real-time sync using Firestore listeners and Firebase Cloud Functions, and **secured** sessions with Firebase Auth across a responsive **Tailwind CSS** UI.

AEVM – Advanced Electronic Voting Machine

GitHub | Live

- **Engineered** a dual-biometric (fingerprint + facial) verification pipeline achieving **100% voter identity accuracy** and eliminating impersonation attempts.
- **Implemented** real-time Firebase synchronization for instant vote logging, **reducing result computation time by 99%**.
- **Developed** a Raspberry Pi IoT hardware stack with a **React + Tailwind** admin dashboard; **published research**: “*AEVM: Enhancing Electoral Integrity with Biometric Verification.*”

EDUCATION

VIT-AP University	2022 – 2026
B.Tech in Electronics & Communication Engineering	CGPA: 8.33
Higher Secondary Education	2021
West Bengal Council of Higher Secondary Education (WBCHSE)	85%
Secondary Education	2019
West Bengal Board of Secondary Education (WBBSE)	89%

PROFESSIONAL CERTIFICATES

Oracle Cloud Infrastructure Foundations Associate – Oracle, July 2025
System Design Masterclass – Udemy, July 2025
Full Stack Developer Bootcamp (Frontend to Backend) – GeeksforGeeks, March 2025
Introduction to Generative AI – Verified Certification
Physical Design & Verification (VLSI) – Entuple Technologies, Aug–Sep 2023

ACHIEVEMENTS

- **Recognized** with a Certificate of Excellence at Codingal Inc. for **2,500+ live sessions** and a **4.44/5** rating (July 2024).
- **Represented** VIT-AP at Great India HackFest 2K24 (CSI Chapter) and V-INN Expo’24 (RGEMS Project category).
- University Finalist at Smart India Hackathon; represented VIT-AP at Mark24 Hackathon.
- Awarded IETE Membership and the Swami Vivekananda Merit-cum-Means Scholarship (2019).
- **Commended** by engineering management at Newton School of Technology for consistently strong student engagement and learning outcomes (March 2026).